

Tucker order-3 vs order-4

node207-23, H100 80GB, BF16 autocast, Llama geometry, seq 1024
16,384 tokens per optimizer step

1 Parameter count

Budget	Layout	Parameters	vs dense
133M	Tucker-4	132,998,816	-124,190,048
133M	Tucker-3: split input	133,002,112	-124,186,752
133M	Tucker-3: split output	132,999,936	-124,188,928
133M	Tucker-3: paired balanced	132,999,680	-124,189,184
225M	Tucker-4	225,023,872	-32,164,992
225M	Tucker-3: split input	225,000,704	-32,188,160
225M	Tucker-3: split output	224,997,120	-32,191,744
225M	Tucker-3: paired balanced	224,999,552	-32,189,312

2 Forward, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	622.9	49.5
133M	Tucker-3: split input	642.6	102.6
133M	Tucker-3: split output	611.9	91.6
133M	Tucker-3: paired balanced	637.8	55.0
225M	Tucker-4	584.8	58.2
225M	Tucker-3: split input	602.5	110.6
225M	Tucker-3: split output	621.5	101.4
225M	Tucker-3: paired balanced	650.6	55.6

3 Backward, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	767.7	87.8
133M	Tucker-3: split input	735.7	149.1
133M	Tucker-3: split output	724.1	173.0
133M	Tucker-3: paired balanced	556.7	73.7
225M	Tucker-4	717.2	113.8
225M	Tucker-3: split input	696.6	174.0
225M	Tucker-3: split output	732.4	203.8
225M	Tucker-3: paired balanced	559.1	76.7

4 Optimizer, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	86.0	66.0
133M	Tucker-3: split input	84.6	68.3
133M	Tucker-3: split output	80.3	57.8
133M	Tucker-3: paired balanced	82.0	55.4
225M	Tucker-4	87.0	71.2
225M	Tucker-3: split input	81.9	72.6
225M	Tucker-3: split output	93.9	88.1
225M	Tucker-3: paired balanced	84.6	66.6

5 Retraction, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	20.9	21.3
133M	Tucker-3: split input	95.2	96.6
133M	Tucker-3: split output	110.3	111.1
133M	Tucker-3: paired balanced	50.4	51.6
225M	Tucker-4	28.1	29.1
225M	Tucker-3: split input	384.2	384.4
225M	Tucker-3: split output	364.4	364.7
225M	Tucker-3: paired balanced	103.6	104.8

6 Full step, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	1500.9	225.4
133M	Tucker-3: split input	1561.2	417.6
133M	Tucker-3: split output	1529.3	434.2
133M	Tucker-3: paired balanced	1329.1	236.9
225M	Tucker-4	1419.6	273.2
225M	Tucker-3: split input	1768.7	742.5
225M	Tucker-3: split output	1818.1	758.8
225M	Tucker-3: paired balanced	1403.1	304.9

7 Forward+backward peak allocated, GB

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	3.09	8.95
133M	Tucker-3: split input	3.07	8.95
133M	Tucker-3: split output	3.07	8.95
133M	Tucker-3: paired balanced	3.75	9.63
225M	Tucker-4	4.21	9.90
225M	Tucker-3: split input	4.19	9.72
225M	Tucker-3: split output	4.20	9.70
225M	Tucker-3: paired balanced	5.34	10.84

8 Full-step peak allocated, GB

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	3.09	8.95
133M	Tucker-3: split input	3.07	8.95
133M	Tucker-3: split output	3.07	8.95
133M	Tucker-3: paired balanced	3.75	9.63
225M	Tucker-4	4.21	9.90
225M	Tucker-3: split input	5.60	9.72
225M	Tucker-3: split output	4.51	9.70
225M	Tucker-3: paired balanced	5.34	10.84

9 Fastest order-3 full step

Budget	MBxAcc	Layout	ms	time/Tucker-4
133M	1x16	Tucker-3: paired balanced	1329.1	0.886x
133M	16x1	Tucker-3: paired balanced	236.9	1.051x
225M	1x16	Tucker-3: paired balanced	1403.1	0.988x
225M	16x1	Tucker-3: paired balanced	304.9	1.116x

10 Run

Field	Value
Commit	1fbad1e
Run	node207-23/gpu6
GPU	NVIDIA H100 80GB HBM3
PyTorch	2.9.1+cu128
CUDA	12.8
Samples	3 warmup + 12 measured

Every order-3 layout has three trainable factors and one fixed singleton buffer. Paired balanced groups three exact input/output factor pairs, pre-packs a BF16 operator, and reuses saved expansion intermediates in backward. Split-input/output leave one side unsplit. Ratios above 1 mean slower than Tucker-4.